Stacking galaxies reveals a universal baryon fountain

Averaging many simulated galaxies turns messy gas flows into a clean, regular inflow-outflow pattern.

arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations

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Background

Galaxies grow by pulling gas in and pushing it back out, a fountain-like cycle that shapes star formation and feedback. Individual systems are too diverse and dynamic to reveal the underlying pattern directly.

> **Observational hints of baryon cycles exist, but no clean, quantitative picture of the full gas-cycling geometry has emerged.**

What this paper is selling

The authors stack gas velocity and property fields across ensembles of simulated galaxies, suppressing case-to-case noise. This exposes strikingly regular fountain patterns and a compact parameterization of the cycle.

Key results to remember

- Stacked velocity fields reveal coherent, fountain-like inflow and outflow across the full ecosystem.
- Outflow cone opening angles and fountain geometry persist across stellar mass and redshift.
- A simple parameterized model reproduces the stacked fields, linking simulations to observations.

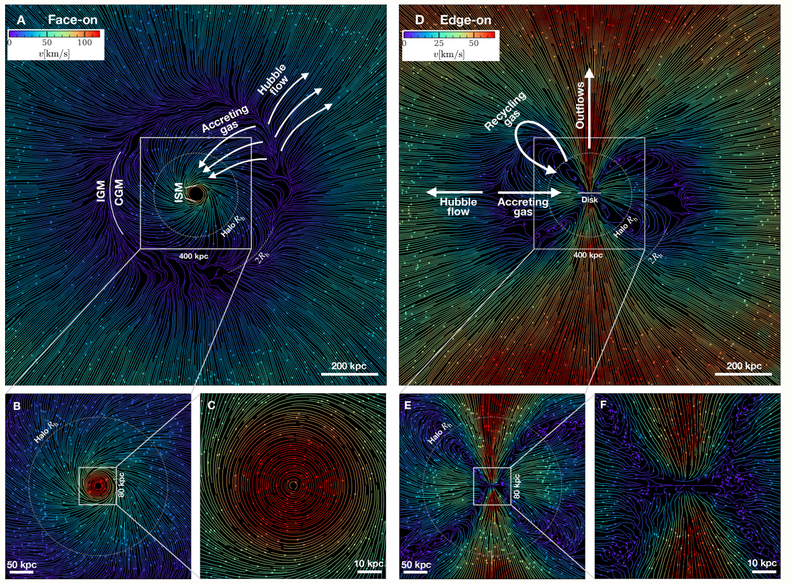


Figure 1 | Stacked gas velocity streamlines show a smooth, coherent fountain pattern in both face-on and edge-on views.

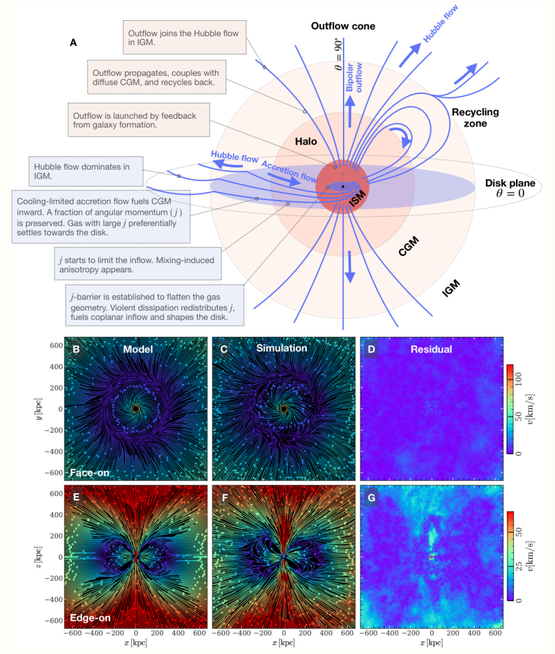


Figure 4 | A schematic fountain model matches the stacked velocity field, with residuals showing how well it captures the cycle.